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# W2 COS30045 Data Visualisation

Lecture

Class

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Agenda

- 2.1 Data visualisation  
Design Guidelines
- 2.2 Visual variables: Marks  
and Channels
- 2.3 Common chart types  
and their use

Agenda

- Data aggregation
- Exploring data with charts

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# W2.3 COS30045

# Lecture

# Agenda

- Common chart types and their use

## Reference

Knaflíc (2015) *Storytelling with data*

- Ch 2



# Common data visualisation types

91%

Simple text



Scatterplot



Vertical bar



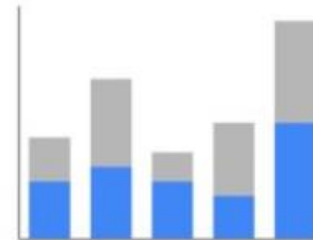
Horizontal bar

|            | A   | B   | C   |
|------------|-----|-----|-----|
| Category 1 | 15% | 22% | 42% |
| Category 2 | 40% | 36% | 20% |
| Category 3 | 35% | 17% | 34% |
| Category 4 | 30% | 29% | 26% |
| Category 5 | 55% | 30% | 58% |
| Category 6 | 11% | 25% | 49% |

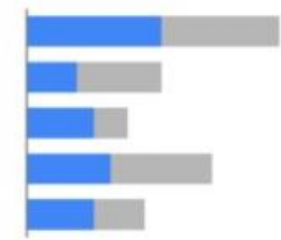
Table



Line



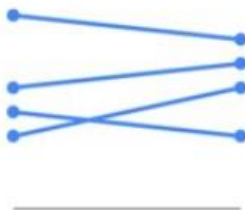
Stacked vertical bar



Stacked horizontal bar

|            | A   | B   | C   |
|------------|-----|-----|-----|
| Category 1 | 15% | 22% | 42% |
| Category 2 | 40% | 36% | 20% |
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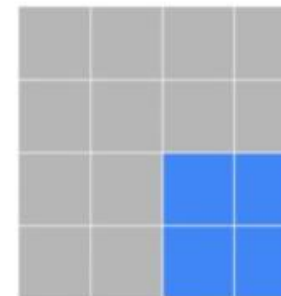
Heatmap



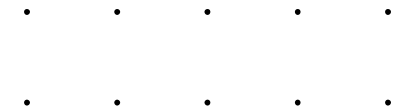
Slopegraph



Waterfall



Square area



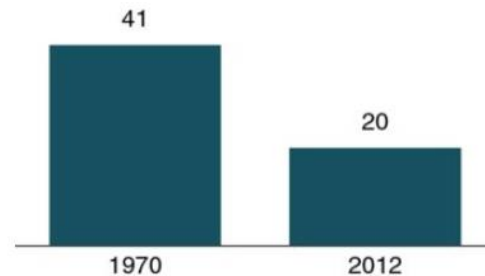
# Common data visualisation types: Text

*When you have just a number or two that you want to communicate: use the numbers directly.*

Knaflic (2015)

## Children with a "Traditional" Stay-at-Home Mother

*% of children with a married stay-at-home mother with a working husband*



Note: Based on children younger than 18. Their mothers are categorized based on employment status in 1970 and 2012.

Source: Pew Research Center analysis of March Current Population Surveys Integrated Public Use Microdata Series (IPUMS-CPS), 1971 and 2013

Adapted from PEW RESEARCH CENTER

# 20%

of children had a  
**traditional stay-at-home mom**  
in 2012, compared to 41% in 1970

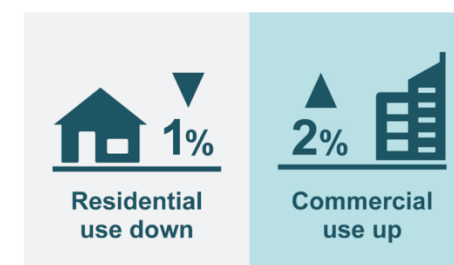
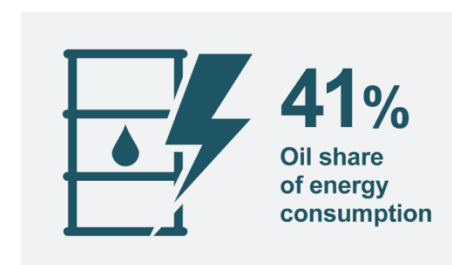
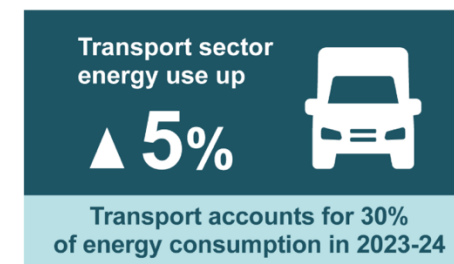
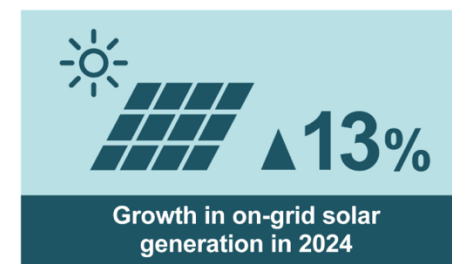
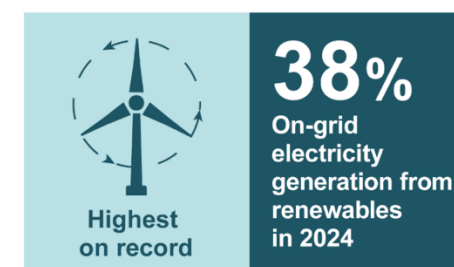
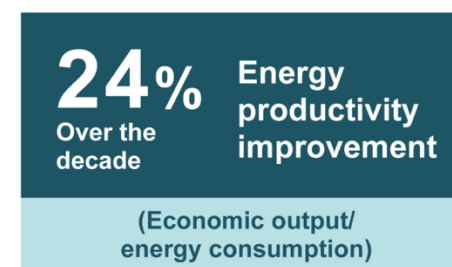
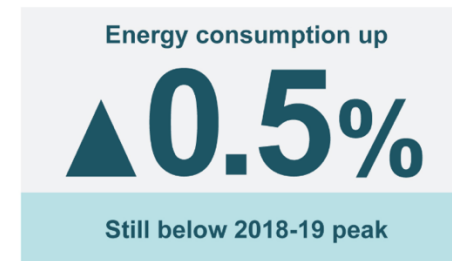
# Example – text, text + infographic

vs charts...

## Data charts



## Australian Energy Statistics In 2023–24



<https://www.energy.gov.au/energy-data/australian-energy-statistics>

# Common data visualisation types: Tables

- Verbal processing (not visual) (read)
- Good for list of attributes with different levels of measurement (nominal, ordinal, interval, ratio)
- Design recommendation: Borders
  - Don't let table borders overwhelm data
  - Use light borders or white space
  - Create separate visual layers for data and table structure

Heavy borders

| Group   | Metric A | Metric B | Metric C |
|---------|----------|----------|----------|
| Group 1 | \$X.X    | Y%       | Z,ZZZ    |
| Group 2 | \$X.X    | Y%       | Z,ZZZ    |
| Group 3 | \$X.X    | Y%       | Z,ZZZ    |
| Group 4 | \$X.X    | Y%       | Z,ZZZ    |
| Group 5 | \$X.X    | Y%       | Z,ZZZ    |

Light borders

| Group   | Metric A | Metric B | Metric C |
|---------|----------|----------|----------|
| Group 1 | \$X.X    | Y%       | Z,ZZZ    |
| Group 2 | \$X.X    | Y%       | Z,ZZZ    |
| Group 3 | \$X.X    | Y%       | Z,ZZZ    |
| Group 4 | \$X.X    | Y%       | Z,ZZZ    |
| Group 5 | \$X.X    | Y%       | Z,ZZZ    |

Minimal borders

| Group   | Metric A | Metric B | Metric C |
|---------|----------|----------|----------|
| Group 1 | \$X.X    | Y%       | Z,ZZZ    |
| Group 2 | \$X.X    | Y%       | Z,ZZZ    |
| Group 3 | \$X.X    | Y%       | Z,ZZZ    |
| Group 4 | \$X.X    | Y%       | Z,ZZZ    |
| Group 5 | \$X.X    | Y%       | Z,ZZZ    |

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Figure 2.4 Table borders

# Common data visualisation types: Heat map

- Table with visual cues
- Use saturation to show **relative** differences between numbers

Table

|            | A   | B   | C   |
|------------|-----|-----|-----|
| Category 1 | 15% | 22% | 42% |
| Category 2 | 40% | 36% | 20% |
| Category 3 | 35% | 17% | 34% |
| Category 4 | 30% | 29% | 26% |
| Category 5 | 55% | 30% | 58% |
| Category 6 | 11% | 25% | 49% |

Heatmap

LOW-HIGH

|            | A   | B   | C   |
|------------|-----|-----|-----|
| Category 1 | 15% | 22% | 42% |
| Category 2 | 40% | 36% | 20% |
| Category 3 | 35% | 17% | 34% |
| Category 4 | 30% | 29% | 26% |
| Category 5 | 55% | 30% | 58% |
| Category 6 | 11% | 25% | 49% |

Note legend

**Figure 2.5** Two views of the same data

# Common data visualisation types: Points

## Scatter plot

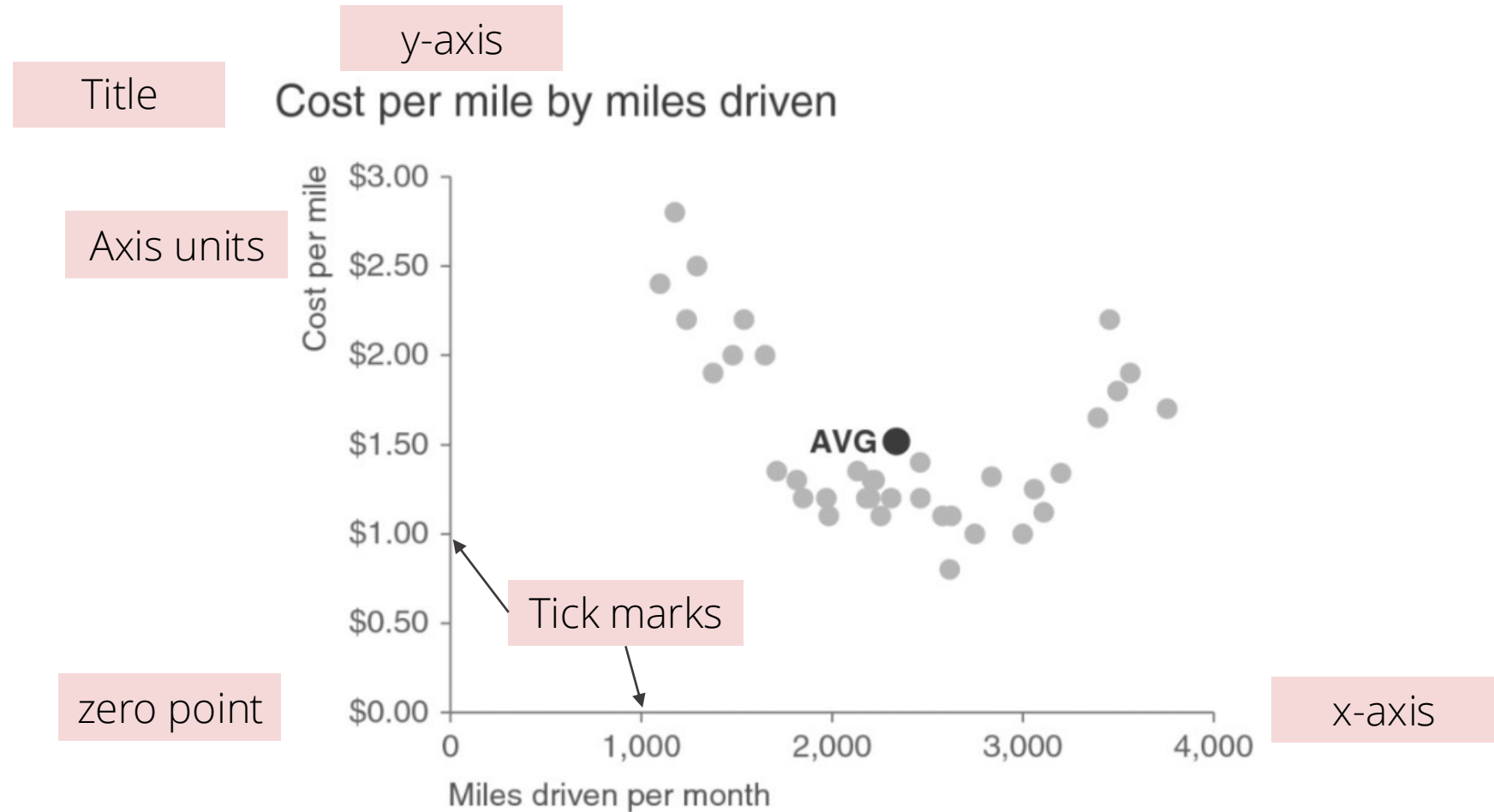


Figure 2.6 Scatterplot

# Common data visualisation types: Points

## Scatter plot

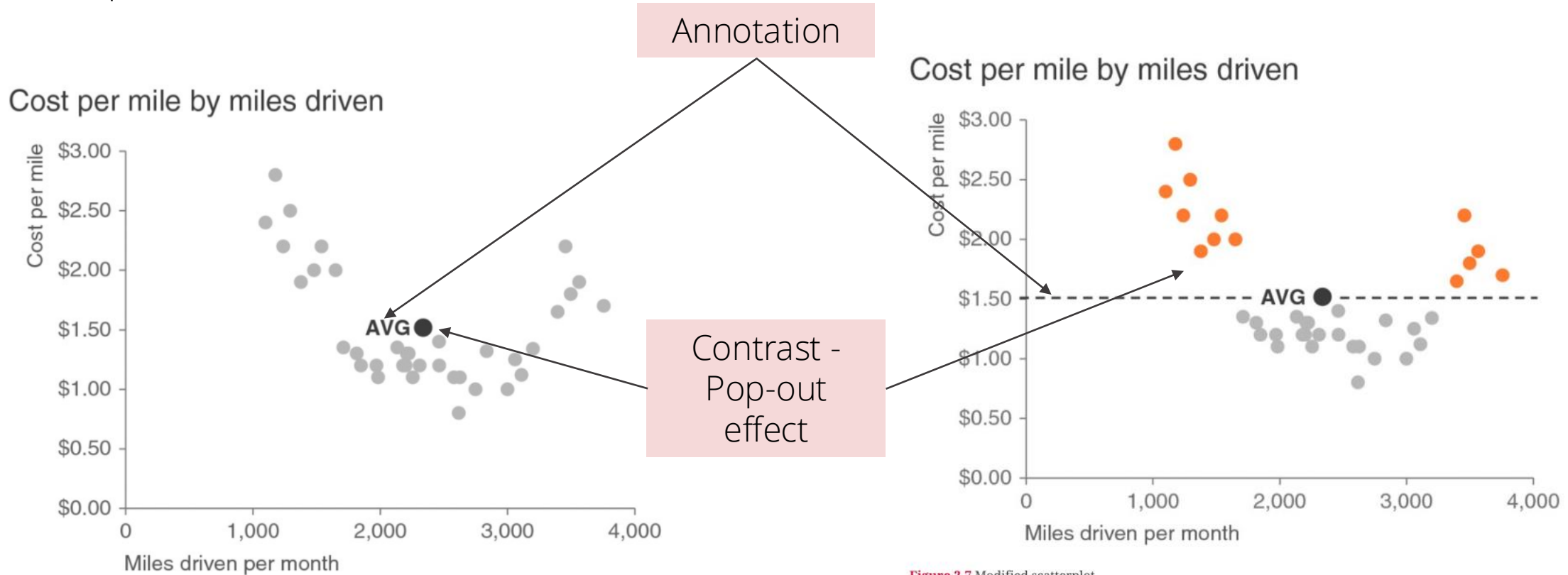
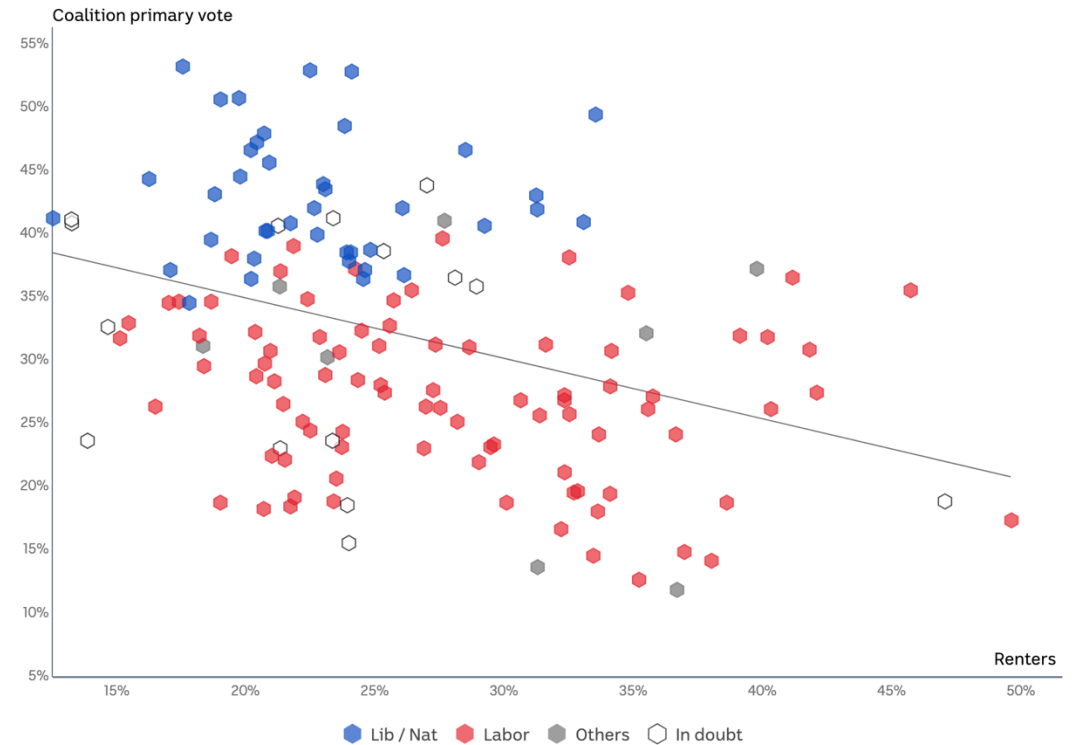
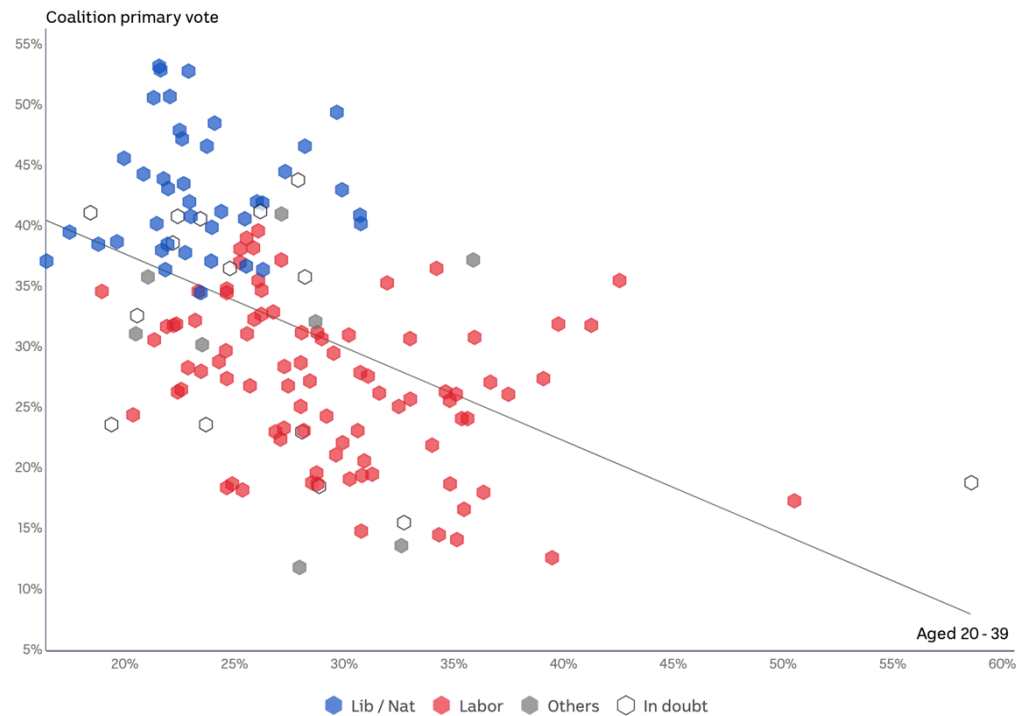


Figure 2.6 Scatterplot

Figure 2.7 Modified scatterplot

# Australian election voting patterns



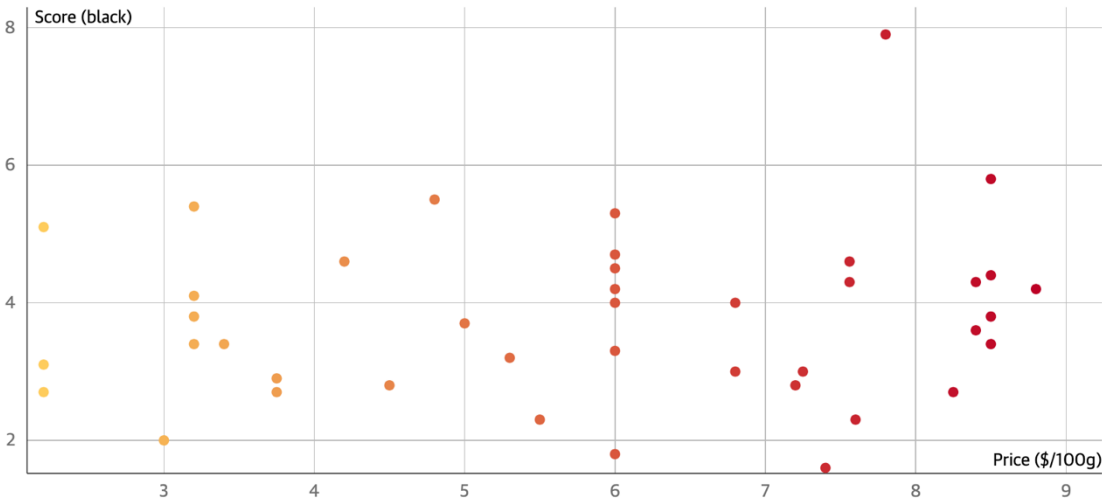
<https://www.abc.net.au/news/2025-05-07/election-result-age-income-housing-trends/105253600>

# Example

## Supermarket coffee taste test results

Showing  v  for 41 different brands

Colour by:

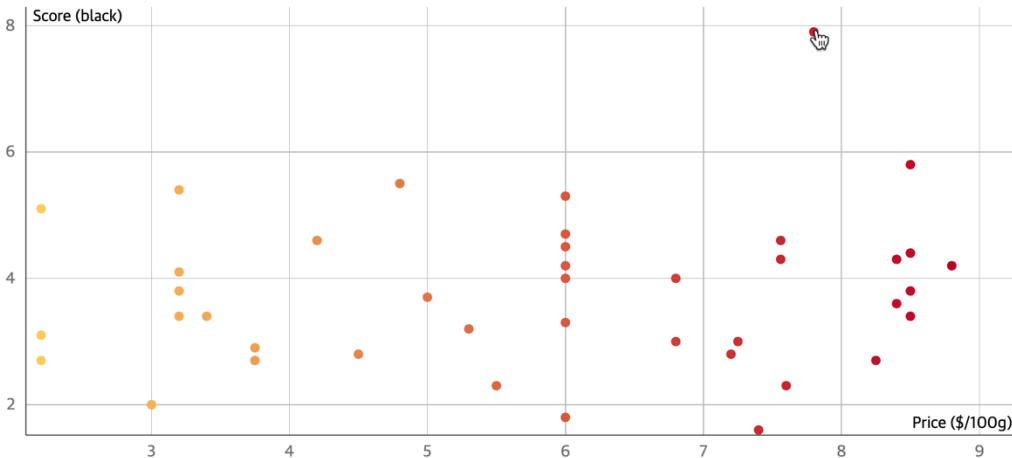


Guardian graphic.

## Supermarket coffee taste test results

Showing  v  for 41 different brands

Colour by:



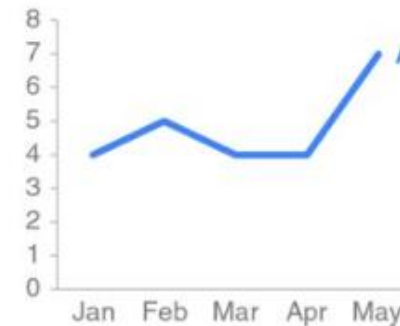
Guardian graphic.

# Common data visualisation types: Lines

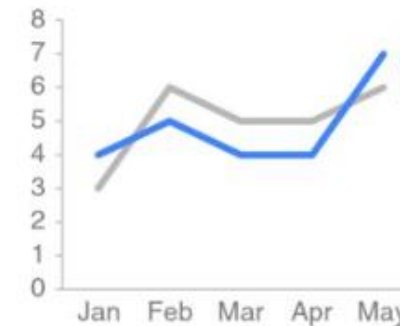
## Line chart

- Use to plot continuous data
- Not suitable for nominal (categorical data)
- Examples,
  - Time (months, days, years)
  - Money
  - Weight, Length
  - Temperature, pressure, humidity

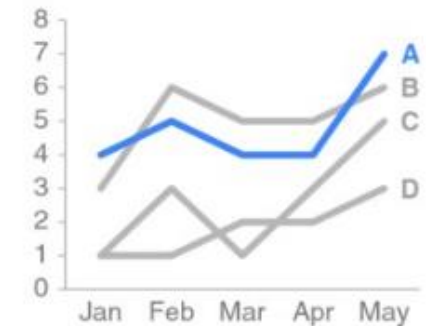
Single series



Two series

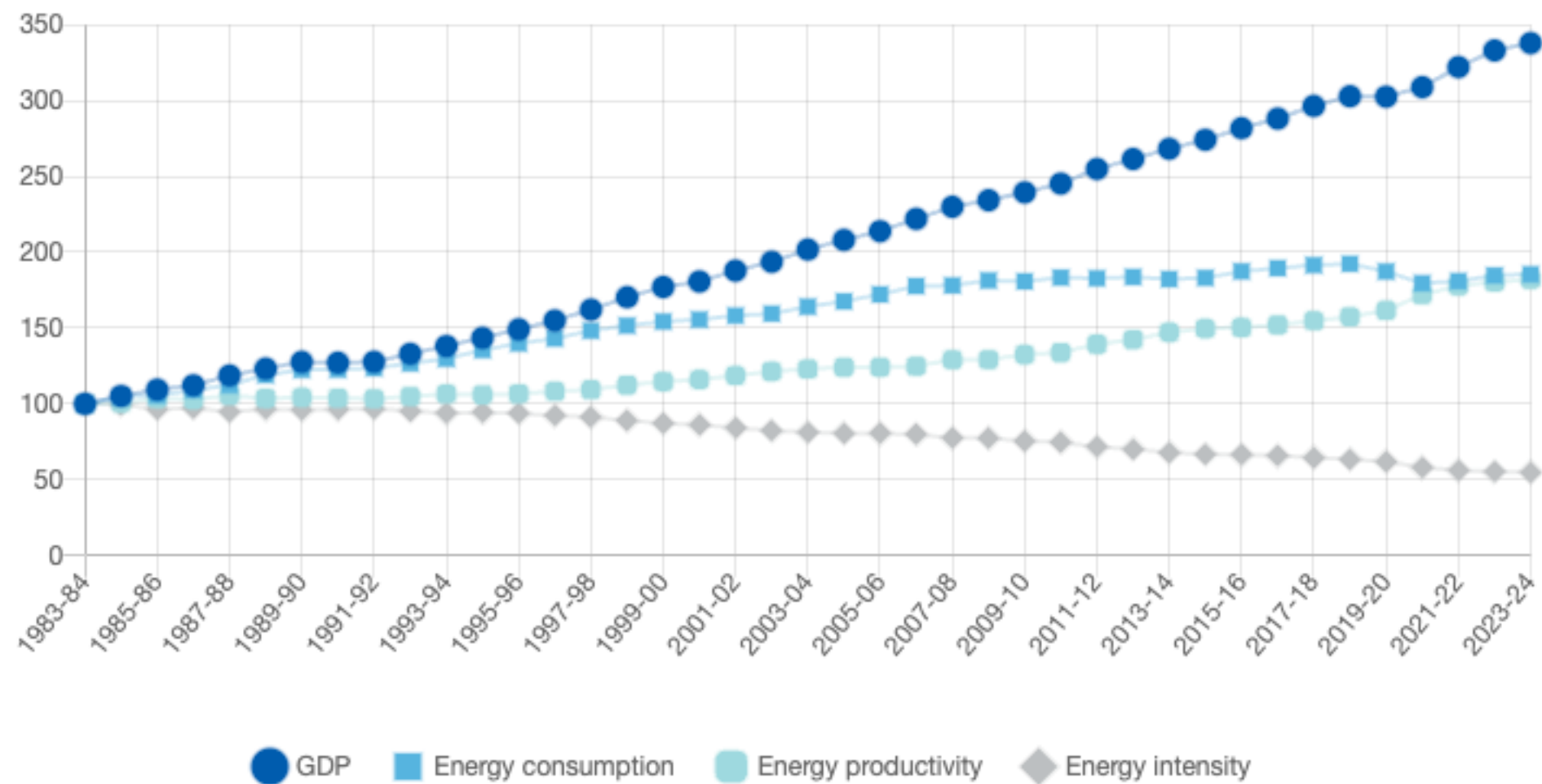


Multiple series

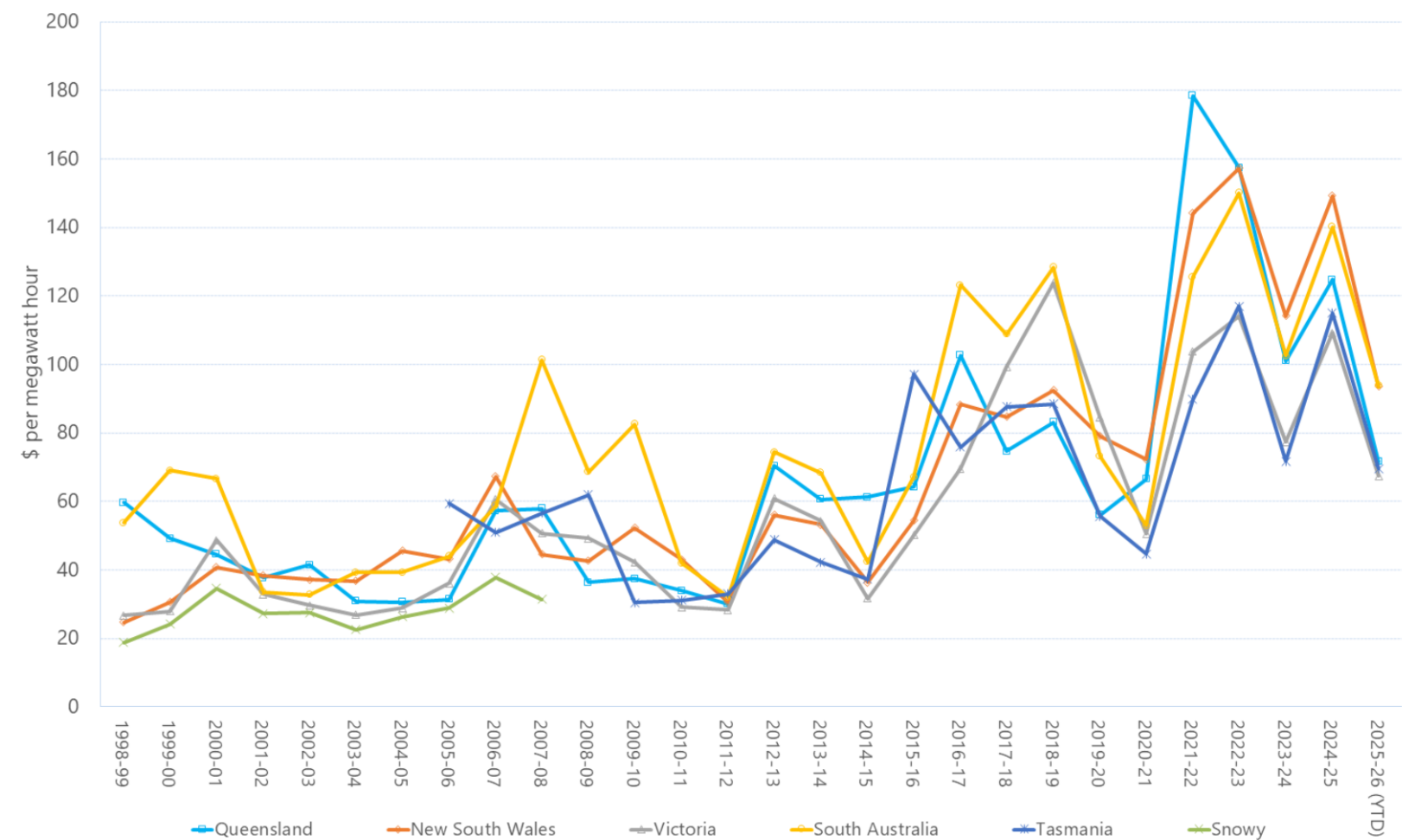


**Figure 2.8** Line graphs

# Example



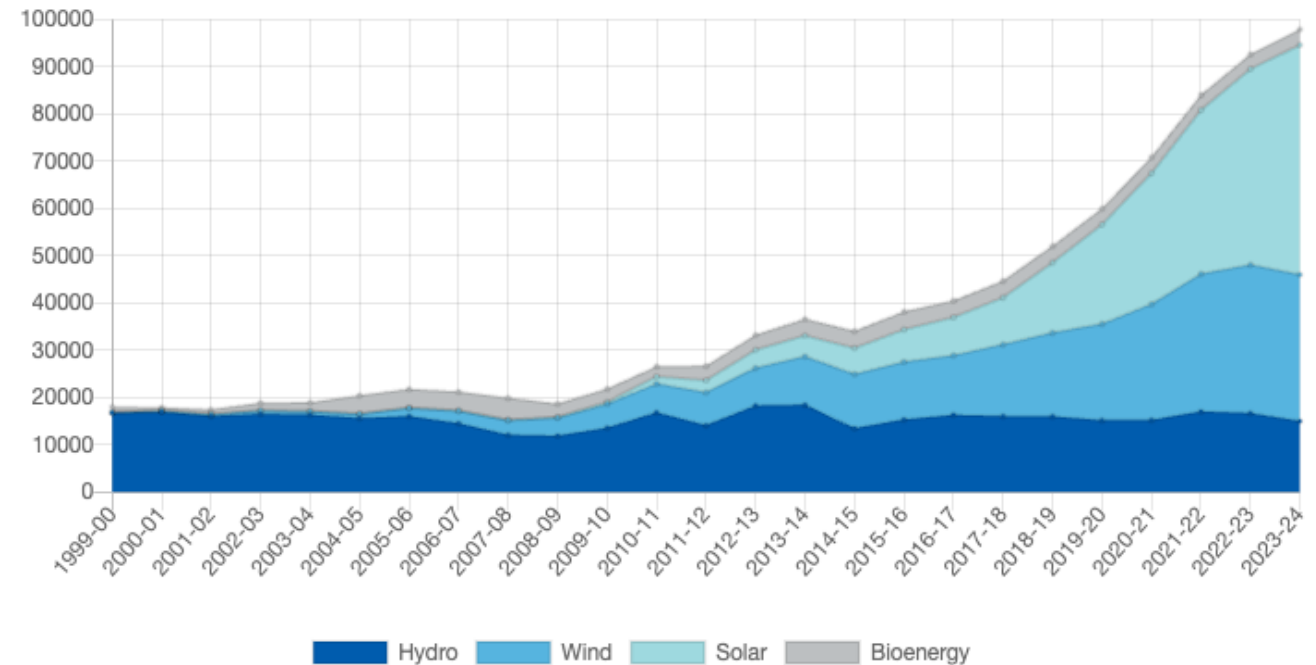
# Electricity price



# Common data visualisation types: Lines

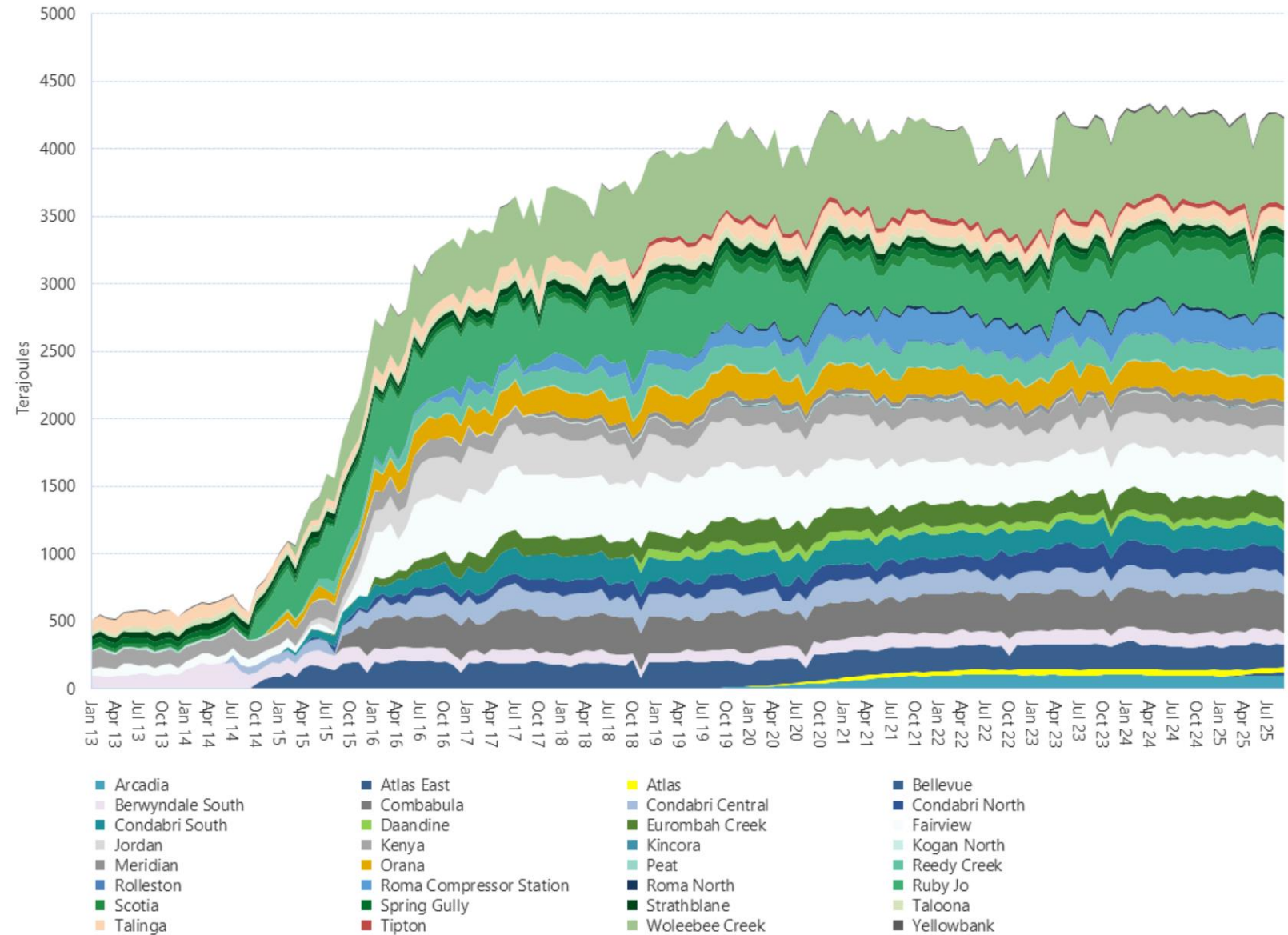
## Stacked line chart:

- Values shown on top of each other
- Each line a cumulative total (year 1 + year 2, + year 3))
- When filled – stacked area chart
- Shows overall trends and relative contributions
- Only bottom line has true baseline:
  - Series 1 = Series 1
  - Series 2 = Series 1 + Series 2...
  - Series 3 = Series 1 + Series 2 + Series 3...
- Can get very cluttered with lots of series



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# Gas Production



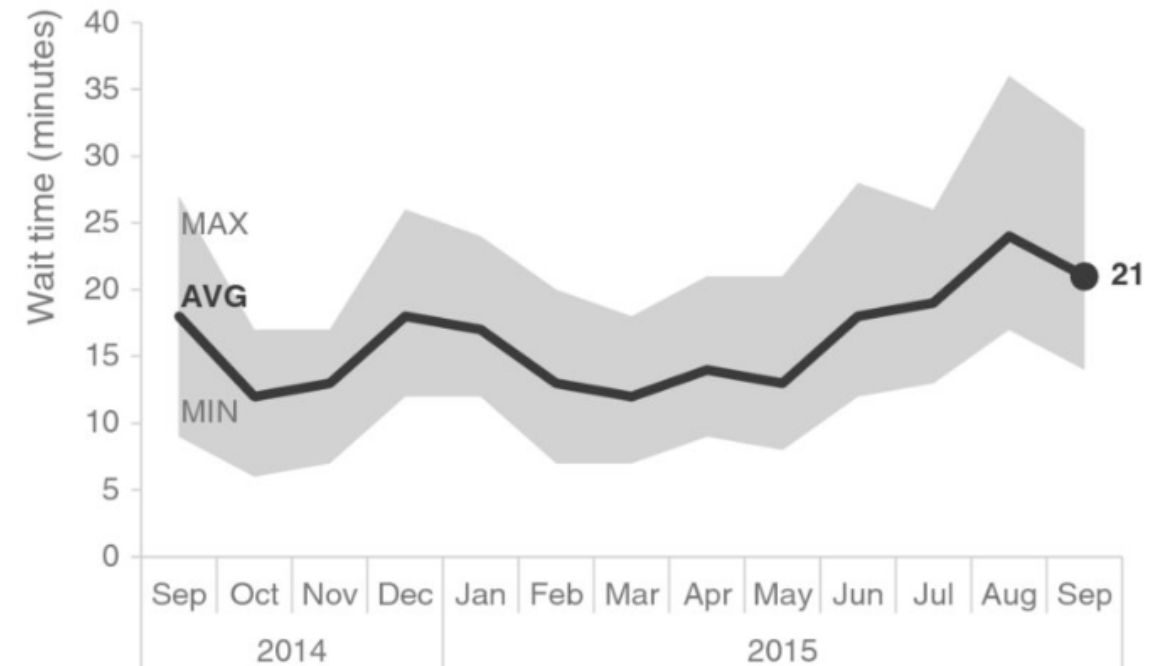
# Common data visualisation types: Lines

## Line graph

- For summary statistics
  - Can plot min and max values when data points are mean values (e.g., average wait time)

### Passport control wait time

Past 13 months

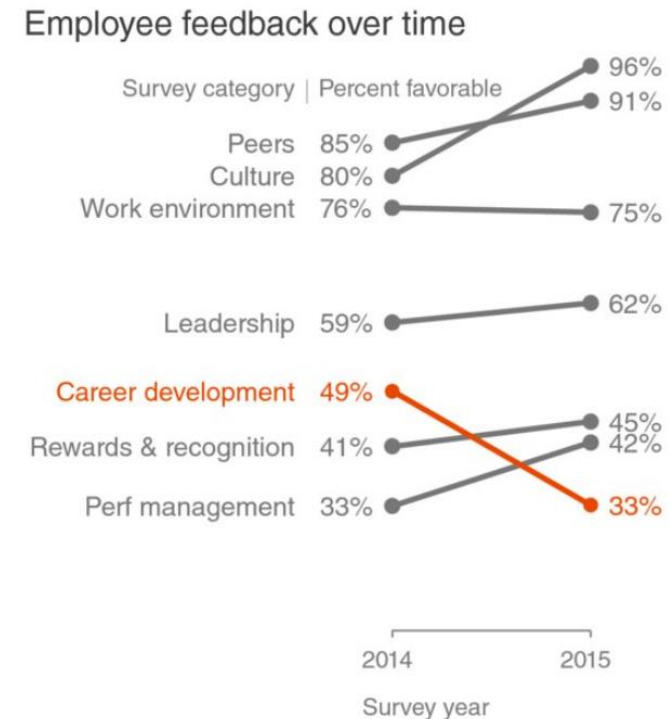
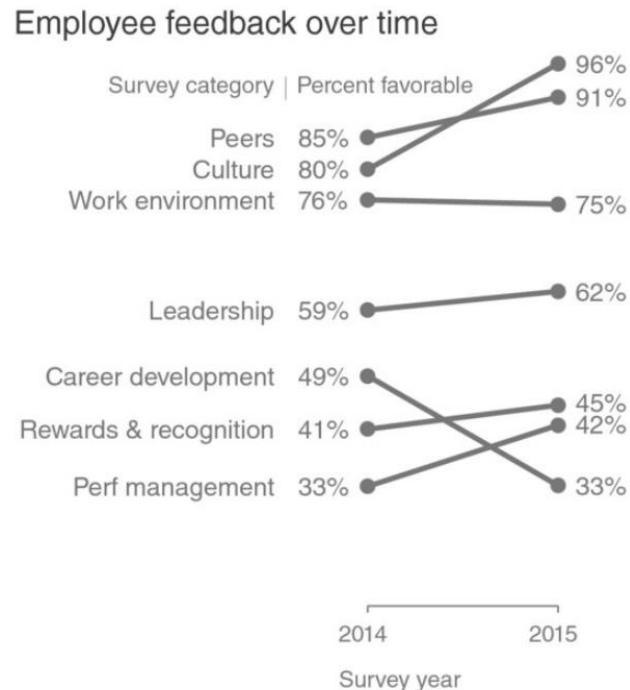


**Figure 2.9** Showing average within a range in a line graph

# Common data visualisation types: Lines

## Slopegraph

- Two or more points comparison (e.g., time 1 vs time 2)
- Show relative increase or decrease



Hint: use annotations to highlight key points of interest

# Common data visualisation types: Bars

## Horizontal bar chart

- Most people familiar with bar charts
- Easy to perceive differences (vertical position)
- Non-zero baseline can distort perception
- Design recommendation:
  - bars should be wider than white space between (Knaflic, 2025)

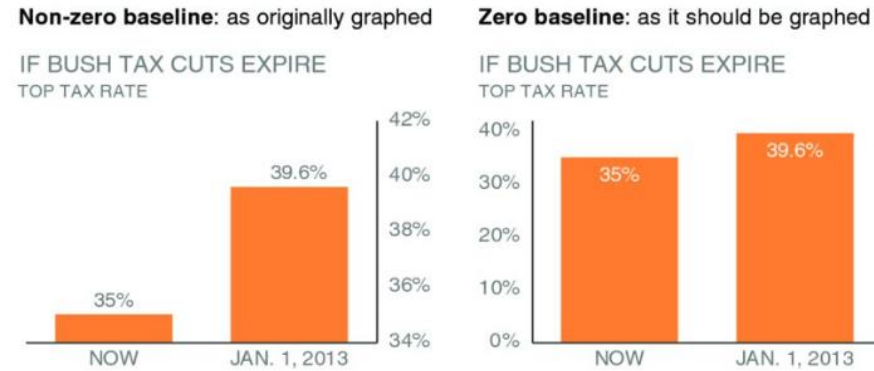
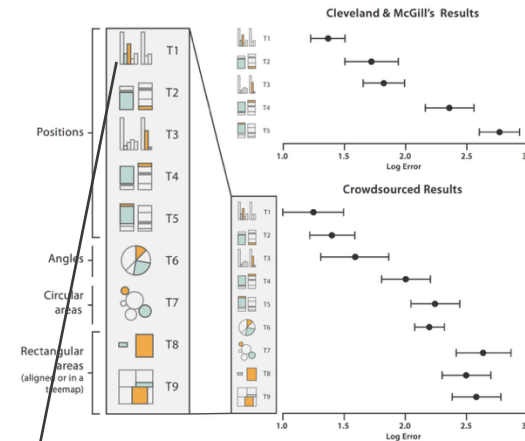


Figure 2.13 Bar charts must have a zero baseline



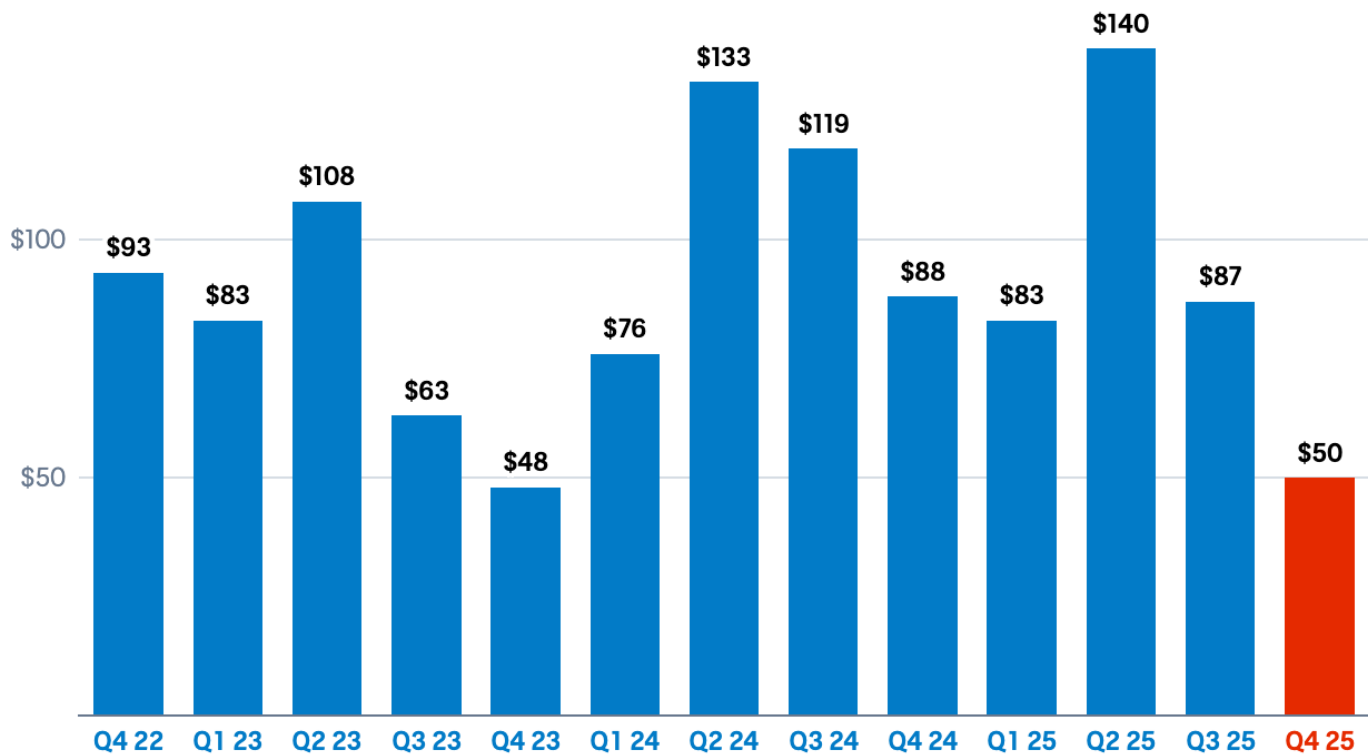
Figure 2.14 Bar width



Position on common scale

# Wholesale electricity prices fell in the last quarter of 2025

Wholesale electricity prices Q4 2025

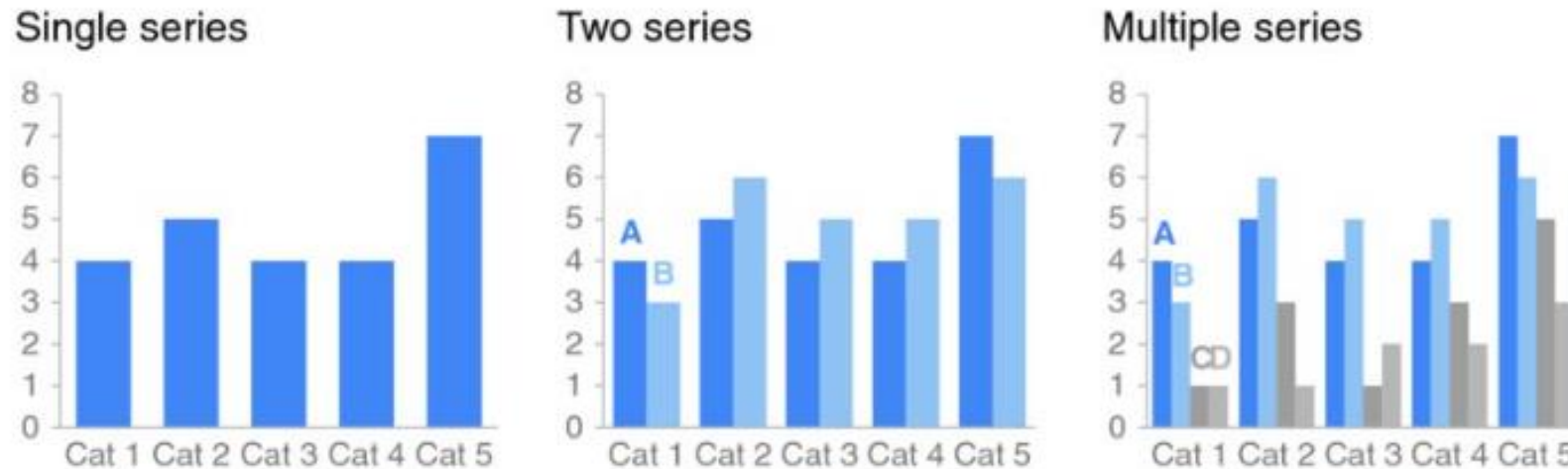


Source: AEMO, quarterly energy dynamics Q4 2025 / [Get the data](#)

# Common data visualisation types: Bar

## Vertical bar chart

- Single or multiple series
- Hard to read with more than 2-3 series (with large number of attributes, consider small multiples (i.e., a set of charts showing subset of data))



**Figure 2.15** Bar charts

# Common data visualisation types: Bar

## Horizontal bar chart

- Easy to read
- Good when category names are long

Design recommendation:

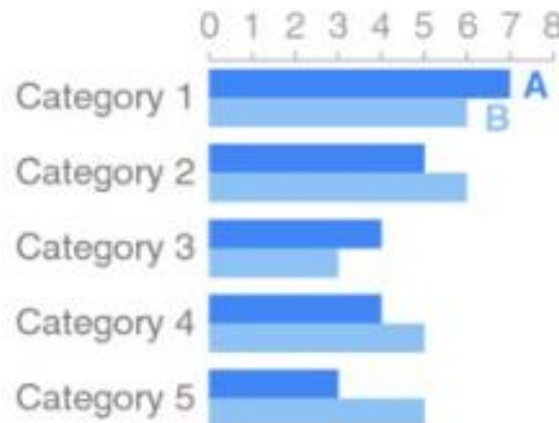
Consider order of categories

- Follow natural order (e.g., high to low)
- Order that helps user interpret data (e.g., sort)

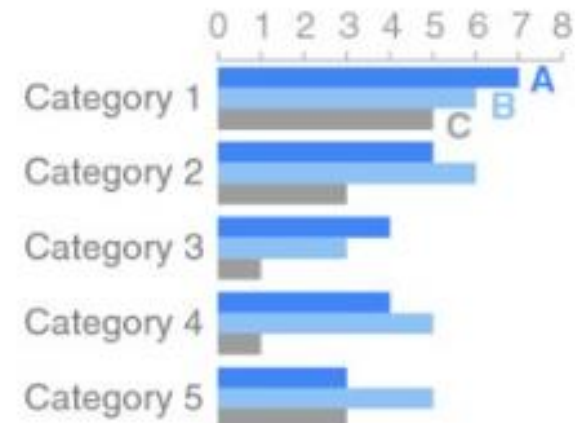
Single series



Two series



Multiple series

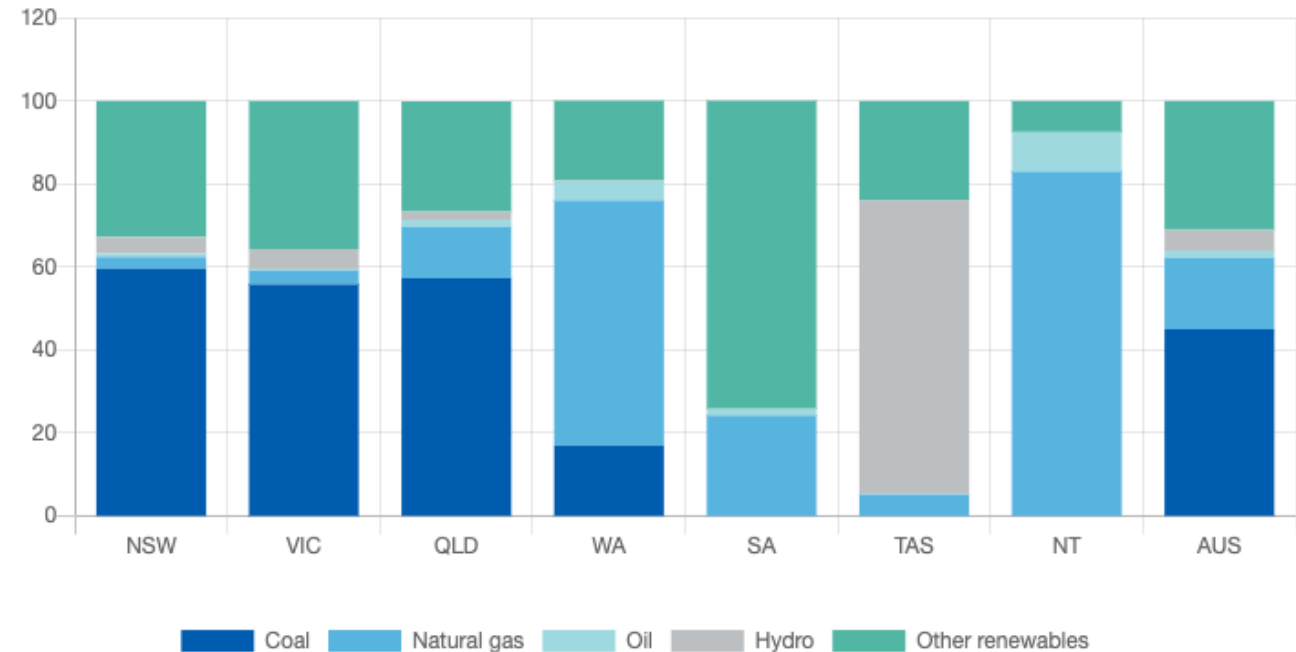


**Figure 2.18** Horizontal bar charts

# Common data visualisation types: Bar

## Proportional stacked bar chart

- Shows relative proportions not absolute numbers
- When pattern more important than exact numbers
- Bars are same height
- Put most important comparison on the bottom



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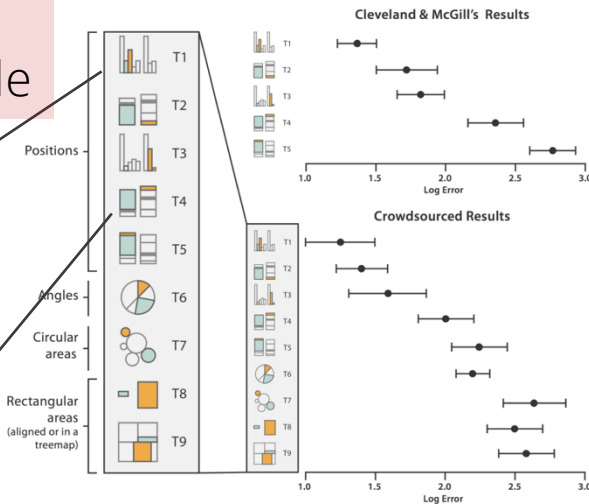
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# Common data visualisation types: Bar

## Stacked vertical bar chart

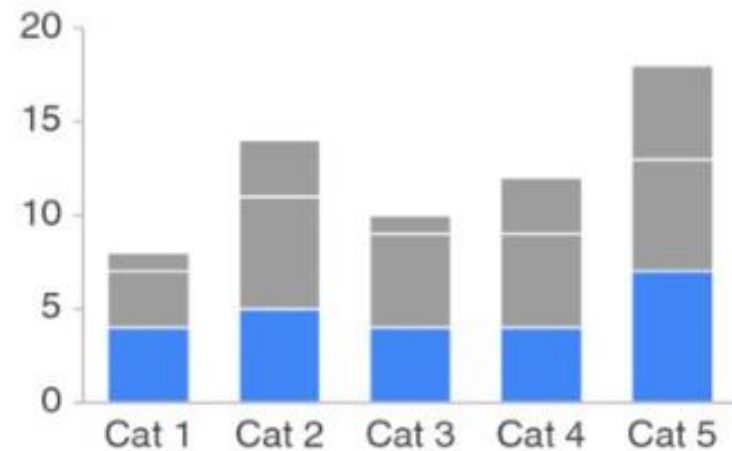
- Compare across categories
- View sub-components within category
- Hard to compare sub-components across categories

Position on common scale

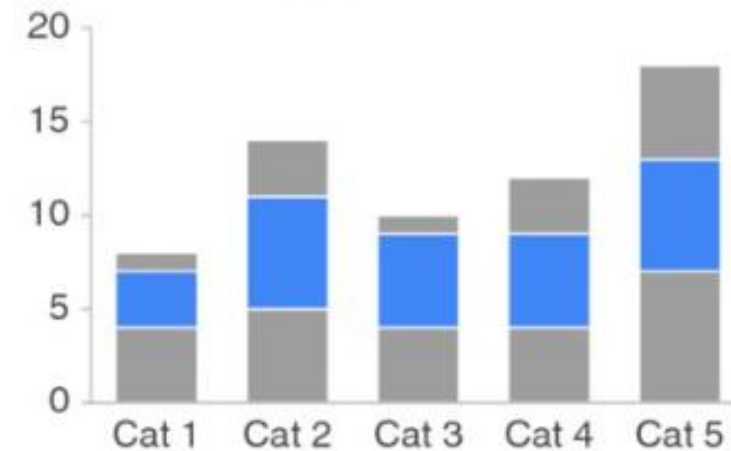


Position on unaligned scale

Comparing **these** is easy



Comparing **these** is hard



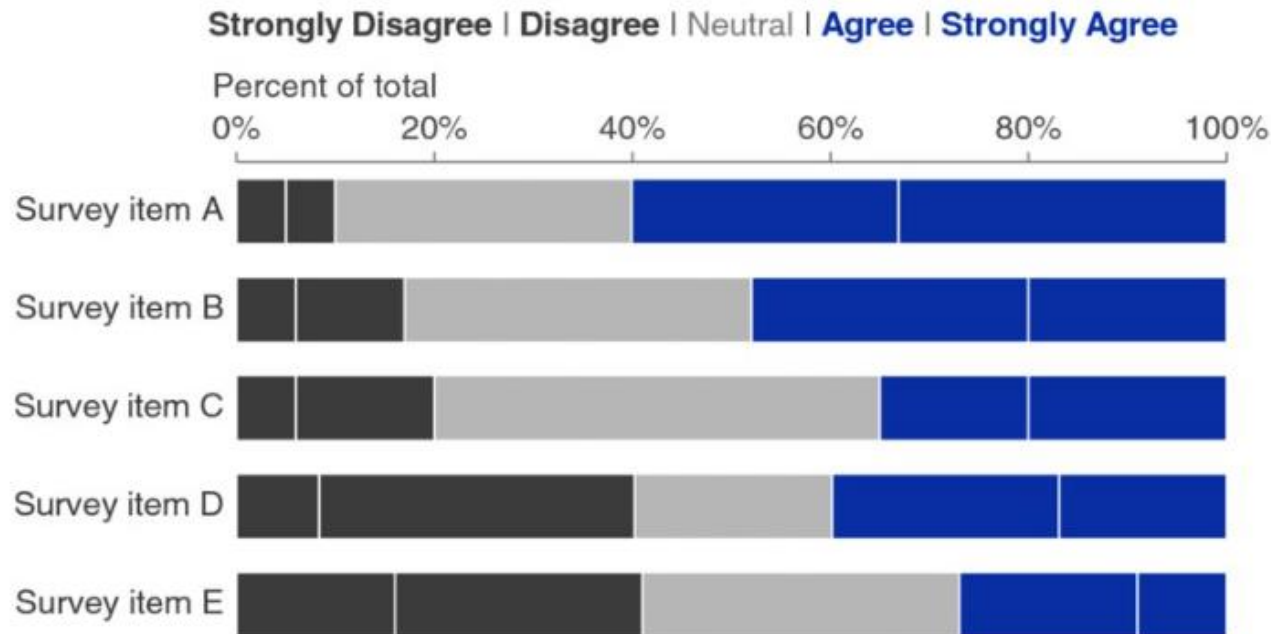
**Figure 2.16** Comparing series with stacked bar charts

# Common data visualisation types: Bar

## Stacked horizontal bar chart

- Similar to stacked vertical bar chart
- Can use absolute or percentage

### Survey results



**Figure 2.19** 100% stacked horizontal bar chart

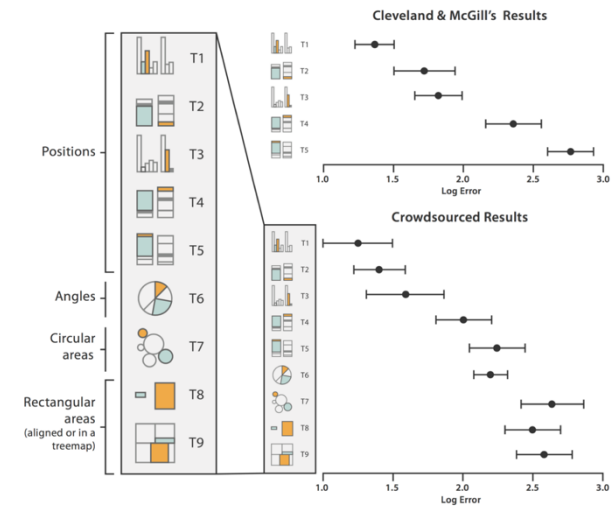
# Common data visualisation types: Area

Difficult to distinguish differences

Design recommendation:

- Only use when large differences between values

Area



## Interview breakdown



Figure 2.20 Square area graph

## Example - Area

# Australian energy trade 2023-24

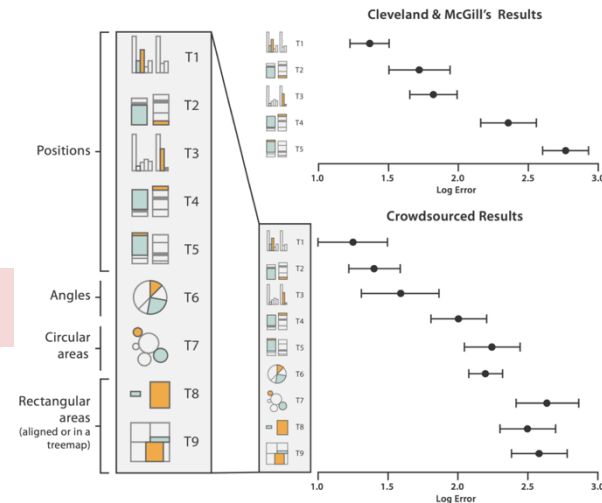


# Common data visualisation types

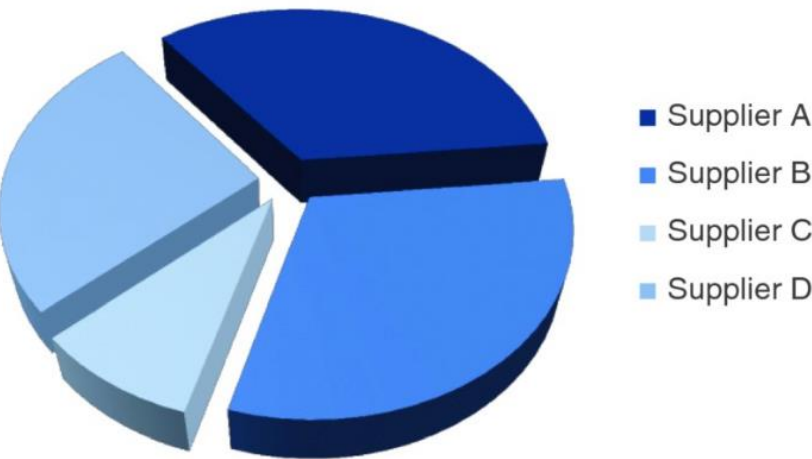
## Pie chart

- Differences in area and angle
- Difficult to judge differences (especially 3D)

Pie



Supplier Market Share



Supplier Market Share

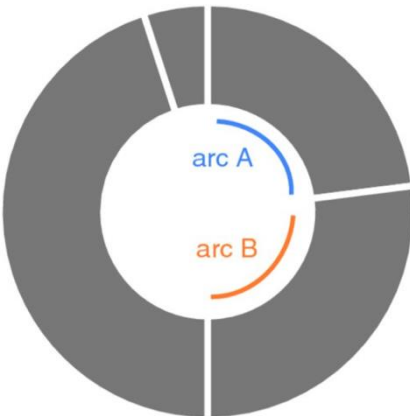
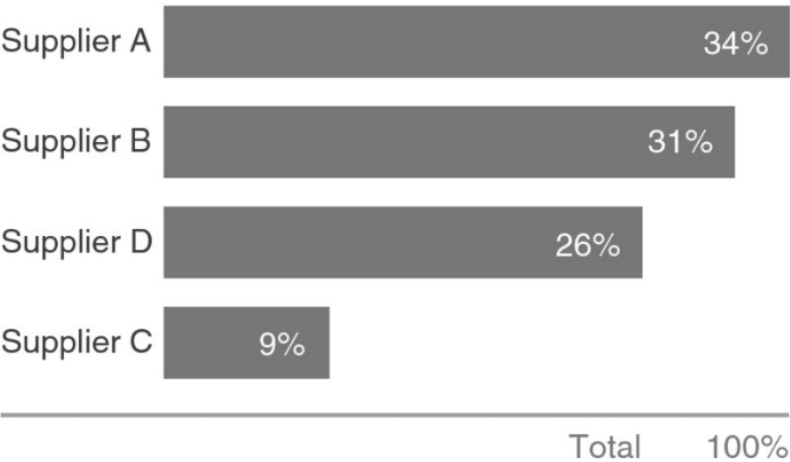
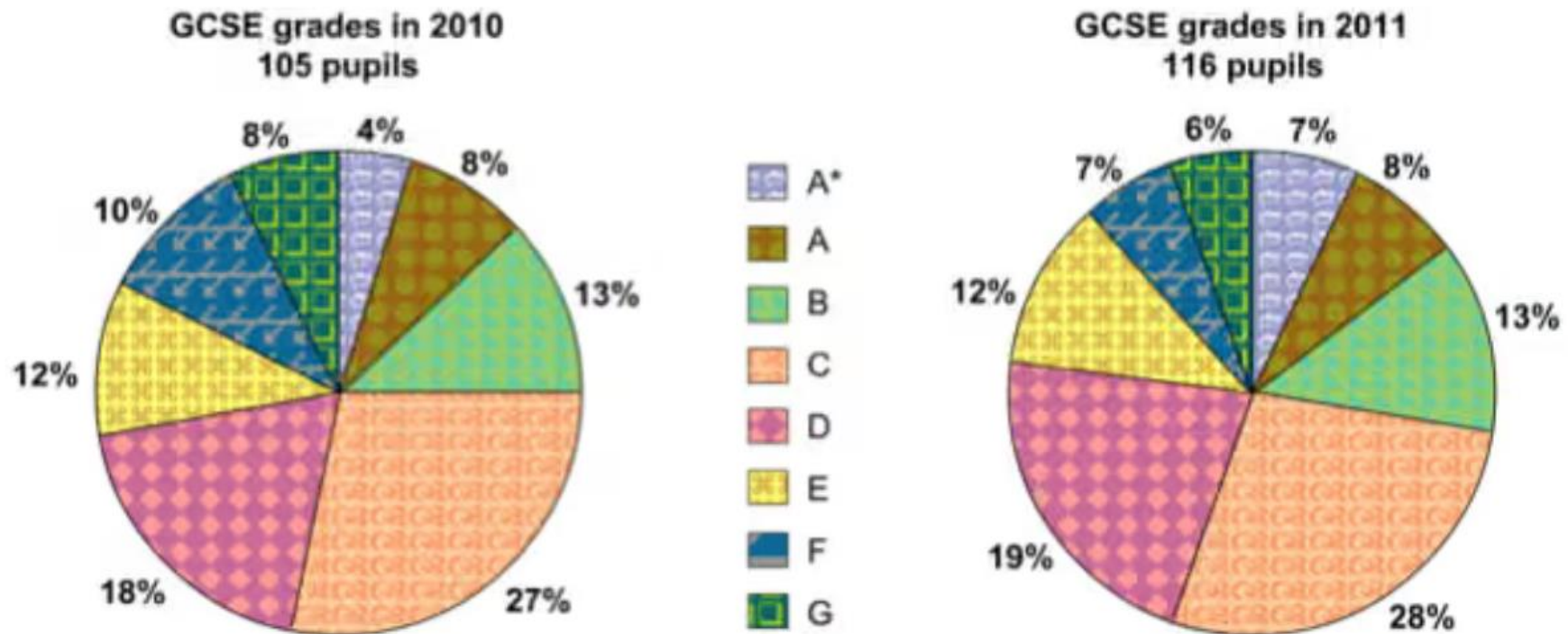


Figure 2.21 Pie chart

Figure 2.23 An alternative to the pie chart

Knaflc (2015)



📷 Pie charts of GCSE results Photograph: Department for Education

<https://www.theguardian.com/teacher-network/2015/nov/22/pass-teachers-numeracy-skills-test-quiz>

# 3D charts

Design recommendation:

- Never use them!

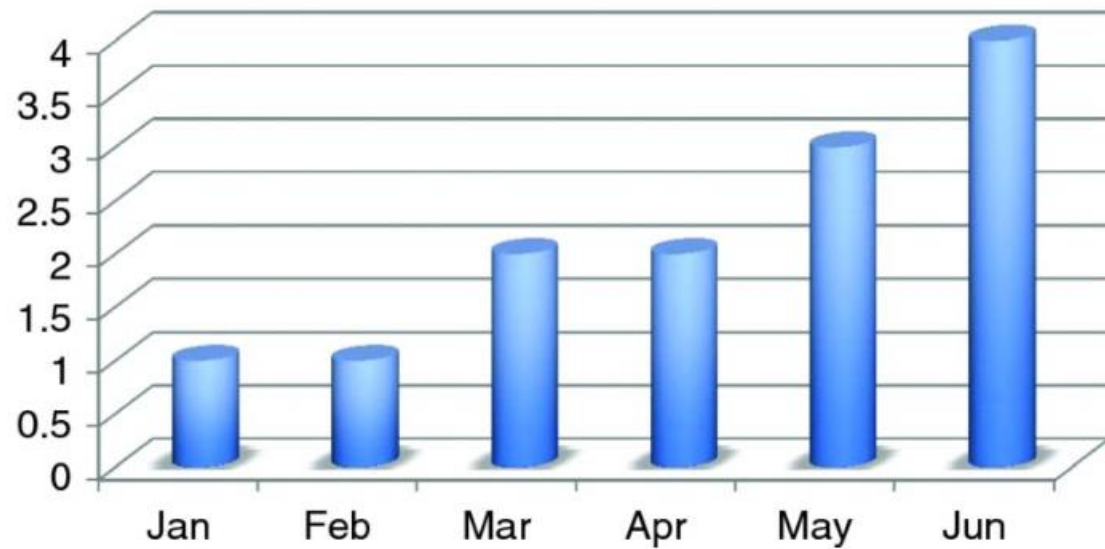


Figure 2.25 3D column chart

# Secondary Axis

Different units plotted on same x-axis

- Can be difficult to work out which axis relates to which plot
- Implies relationship that may not exist

Design recommendation:

- Label second y-axis directly
- Pull apart vertically



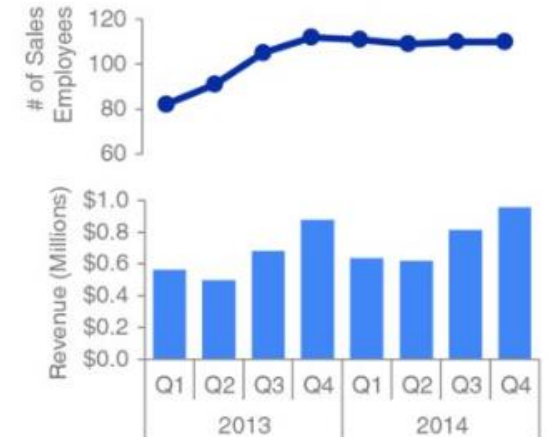
Figure 2.26 Secondary y-axis

Alternative 1: label directly



Figure 2.27 Strategies for avoiding a secondary y-axis

Alternative 2: pull apart vertically



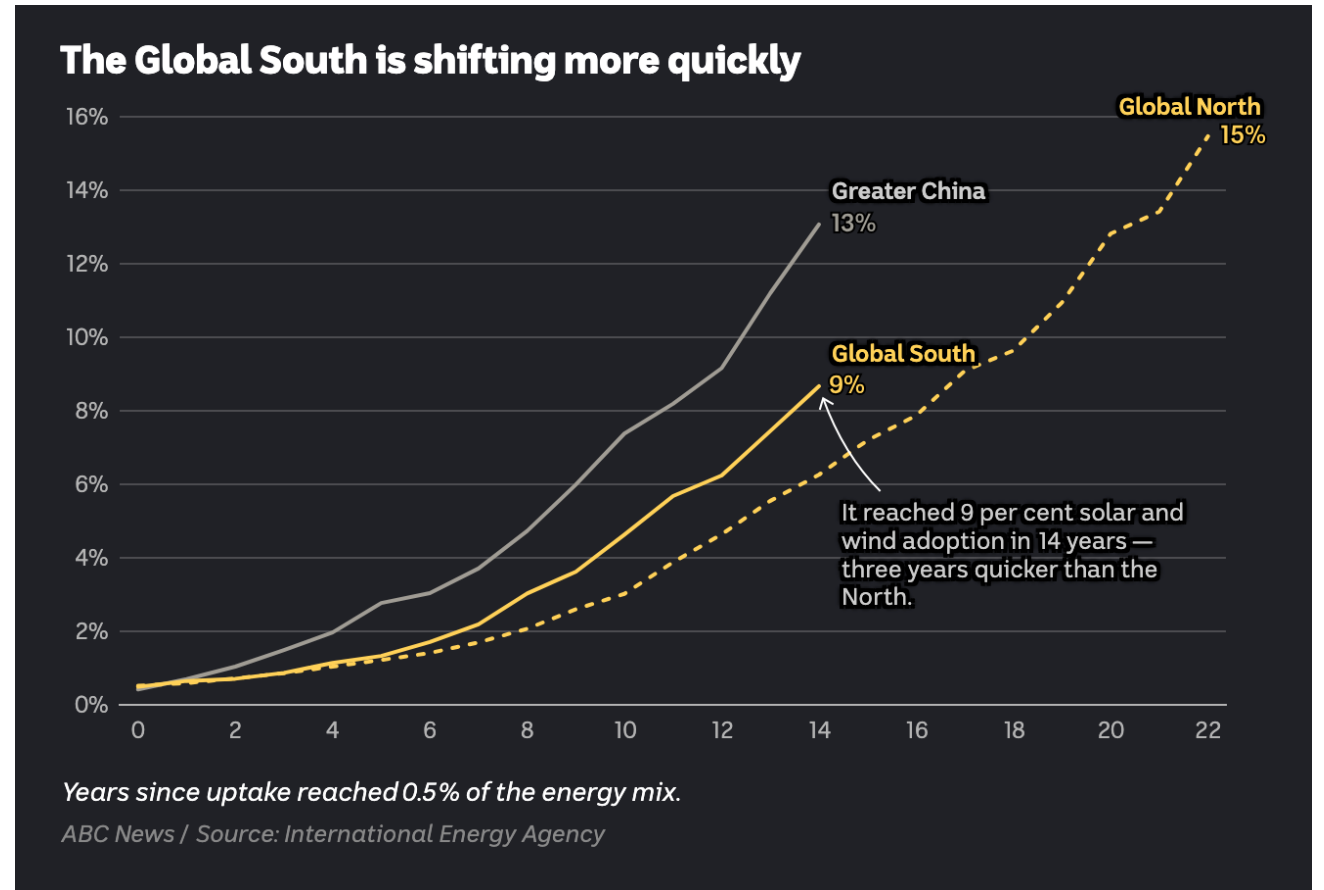
# Summary

Reviewed use of:

- Scatter plots
- Line charts
- Bar charts
- Area and pie charts
- 3D charts
- Multiple axis

Challenge, choose chart that

- shows your data story
- audience easily understands



[https://www.abc.net.au/news/2024-12-18/survey-of-the-worlds-solar-shows-global-boom/10606093?stream=sources:abc-news\\_app&utm\\_medium=content\\_shared&utm\\_campaign=solar\\_news\\_app&utm\\_source=abc](https://www.abc.net.au/news/2024-12-18/survey-of-the-worlds-solar-shows-global-boom/10606093?stream=sources:abc-news_app&utm_medium=content_shared&utm_campaign=solar_news_app&utm_source=abc)